LibTrial — Evidence-Backed Investigation Report

Automated Python Library Evaluation & Sandbox Trial Audit | *Generated: 2026-10-12 09:00:00 UTC*

1. Target Requirement & Project Profile

Project Name	fast-analytics	**Target Python**	`>=3.11`
Total Python Files	1	**Frameworks**	fastapi, pydantic
Requirement	High-performance CSV and Parquet reader with lazy evaluation	**Dependencies**	4 declared

2. Candidate Packages & PyPI Registry Verification

Package	PyPI Status	Version	Requires-Python	License	Dependencies	Compatibility Verdict
`polars`	**Verified**	`v1.24.0`	`>=3.9`	MIT License	2 pkgs	**COMPATIBLE WITH AVAILABLE EVIDENCE**
`duckdb`	**Verified**	`v1.2.0`	`>=3.8`	MIT License	0 pkgs	**COMPATIBLE WITH AVAILABLE EVIDENCE**
`dask`	**Verified**	`v2024.12.0`	`>=3.10`	BSD License	2 pkgs	**COMPATIBLE WITH AVAILABLE EVIDENCE**

3. Gemma 4 Candidate Hypotheses (AI Suggestions — Unverified Facts)

Candidate	Expected Import	Proposed Capability	Fit Rationale
`polars`	`polars`	High-performance DataFrame library with dedicated Lazy API for query optimization and efficient CSV/Parquet reading.	Directly addresses fast analytics requirement with lazy streaming engine.
`duckdb`	`duckdb`	In-process analytical database engine with vectorized SQL and Parquet execution.	In-process SQL querying over CSV and Parquet files.
`dask`	`dask`	Parallel computing library with task graphs for large-scale CSV/Parquet datasets.	Scaling beyond single-machine memory constraints.

4. Isolated Container Integration Trial

Trial Candidate	`polars`	**Outcome**	**PASS**
Exit Code	`0`	**Runtime**	1.85s
Isolation Boundary	Ephemeral Docker Container (--network none)	**Host Execution**	**Zero Host Code Run**

Findings: Process exited normally with returncode 0.; Minimal DataFrame evaluation passed.

5. Safety Boundaries & Audit Limitations

• **Hypothesis Separation:** Model suggestions represent initial unverified hypotheses; only registry data from PyPI and container execution status represent empirical facts.
• **Dependency Completeness:** PyPI metadata only reflects declared direct dependencies; transitive sub-dependencies are not fully SAT-solved.

• **Host Safety:** All trial probes are executed within disposable, network-isolated Docker containers or aborted if Docker is absent. No candidate code was executed on the host system.

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